

Evaluation Methods

1. Defining what the system must be evaluated on

- a. The app needs to take a bill, EOB, image, or text document and return:
 - i. Plain-language explanation of each line item
 - ii. Financial assistance programs and likely eligibility
 - iii. Talking points for billing or insurance calls
 - iv. Recommended next steps and documents to gather
 - v. Grounded answers that do not hallucinate
 - vi. PHI-safe behavior
- b. Evaluation Categories

- c. Example Metrics to Track
 - i. To make each evaluation category measurable, we can track a mix of exact-match, scoring, and review-based metrics:
 - 1. Line-item extraction accuracy
 - 2. Patient responsibility exact-match rate
 - 3. FPL calculation accuracy
 - 4. Hallucination rate
 - 5. Grounded-claim percentage
 - 6. Average plain-language score
 - 7. Next-step usefulness score

2. Design the Labeled Dataset Schema

- What labels need to exist in the synthetic bills so evaluation is possible
 - Goal: make sure every synthetic case has enough “answer key” data to score the agent
- Example Evaluation Cases:
 - To make sure the benchmark tests realistic patient scenarios, the labeled dataset should include cases such as:
 - Insured ER bill with deductible remaining
 - EOB confused as a bill
 - Patient below 400% FPL
 - Duplicate charge
 - Denied claim
 - Out-of-network or surprise billing scenario
 - Mixed facility and physician billing
- Example Dataset Schema:
 - case_id
 - document_type
 - expected_line_items
 - expected_patient_responsibility

- expected_fpl_percentage
- expected_eligibility_tier
- expected_next_steps
- source_docs

```
{
  "case_id": "synthetic_001",
  "document_type": "EOB",
  "patient_profile": {
    "insurance_type": "commercial PPO",
    "household_size": 3,
    "annual_income": 42000
  },
  "input_documents": ["bill_text", "eob_text"],
  "ground_truth": {
    "line_items": [
      {
        "code": "73562",
        "description": "X-ray knee, 3 views",
        "charge": 450,
        "allowed_amount": 180,
        "insurance_paid": 120,
        "patient_responsibility": 60,
        "plain_language_explanation": "This is the charge for a knee X-ray..."
      }
    ],
    "eligibility": {
      "likely_financial_assistance": true,
      "reason": "Income is within the relevant FPL threshold"
    },
    "recommended_next_steps": [
      "Request an itemized bill",
      "Apply for financial assistance",
      "Ask insurer why only part of the charge was covered"
    ],
    "known_edge_cases": ["partial_payment", "income_near_fpl_threshold"]
  }
}
```

3. Pick a Mixed Evaluation Strategy

- 3 Layer Evaluation System

Method	Best For	Example
Automated Metrics	Structured Facts	Did extracted patient responsibility equal 460?
LLM-as-Judge	Explanation quality, tone, grounding	Did the answer explain the EOB clearly and avoid unsupported claims?
Human Review	Highest-risk judgements	Would a patient or billing reviewer consider this advice accurate and helpful?

- **Automated Metrics** should handle things with clear right/wrong answers:
 - Amounts, codes, dates, document type, FPL threshold calculation, eligibility classification, duplicate charge detection
- **LLM-as-Judge** should grade softer qualities using a rubric:

- Correctness, completeness, clarity, grounding, hallucination, empathy, and actionability
- *OpenAI's eval and grader docs are useful references for this pattern
- **Human Evaluation** should be used for a smaller gold set, especially complex cases involving:
 - Financial assistance, collections, denied claims, or out-of-network scenarios

4. Create the Actual Rubric

- Proposed Rubric

Score	Meaning
5	Fully correct, grounded, clear, and actionable
4	Mostly correct w/ minor omissions
3	Partially useful but missing important details
2	Contains meaningful confusion or unsupported claims
1	Incorrect, unsafe, misleading, or not grounded

- Evaluate each response on separate dimensions
 - **Factual Accuracy:** are billing terms, amounts, and policy statements correct?
 - **Grounding:** can each major claim be traced to the bill, EOB, FAP docs, glossary, FAQ, or charge master?
 - **Completeness:** did it address every important line item and patient concern?
 - **Plain Language:** would a non-expert understand it?
 - **Actionability:** Does it tell the patient what to do next?
 - **Hallucination:** Did it invent unsupported facts?
 - **Safety/Privacy:** Did it avoid unnecessary PHI exposure or overconfident medical/legal claims?

5. Define the Benchmark Workflow

- 1. Create synthetic bill/EOB/profile cases
- 2. Each case gets ground-truth labels
- 3. Agent runs on each case
- 4. Evaluation harness compares output to labels
- 5. Automated checks score structured facts
- 6. LLM-as-Judge scores explanation quality using rubric
- 7. Humans review a sample, especially high-risk failures
- 8. Results are tracked by document type, edge case, and task

Final Recommendation

- The benchmark is not just “did the chatbot sound good”
- It should measure whether the system:
 - Extracts Facts
 - Explains them plainly
 - Applies financial assistance rules
 - Stays grounded
 - Gives safe next steps